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Pseudocode for IDA* Search Algorithm

Aim

To write the pseudocode for the IDA* (Iterative Deepening A*) search algorithm.

Idea

IDA* combines the memory saving of iterative deepening DFS with the heuristic guidance of A*. Instead of a depth limit, it uses a limit on the cost f(n) = g(n) + h(n), where g(n) is the cost from the start to n and h(n) is the estimated cost from n to the goal. Each round is a depth first search that stops at any node whose f value is above the current threshold. If the goal is not found, the threshold is raised to the smallest f value that went over the old threshold, and the search starts again.

Pseudocode

```
function IDA_STAR(start, goal)
    threshold <- h(start)
    path <- [start]
    loop
        result <- SEARCH(path, 0, threshold)
        if result = FOUND then
            return path
        if result = INFINITY then
            return FAILURE            // no solution exists
        threshold <- result           // smallest f that exceeded old limit
    end loop
end function

function SEARCH(path, g, threshold)
    node <- last node of path
    f <- g + h(node)
    if f > threshold then
        return f                      // cut off, report this f value
    if node = goal then
        return FOUND
    min <- INFINITY
    for each successor s of node do
        if s is not in path then      // avoid cycles
            add s to end of path
            result <- SEARCH(path, g + cost(node, s), threshold)
            if result = FOUND then
                return FOUND
            if result < min then
                min <- result
```

```
            remove last node from path
    end for
    return min
end function
```

Explanation of the steps

1. The first threshold is the heuristic value of the start node, h(start).
2. SEARCH goes depth first. A node is expanded only if f = g + h is not more than the threshold.
3. If f is more than the threshold, the node is cut off and its f value is returned.
4. After exploring all successors, SEARCH returns the smallest f value it saw above the threshold.
5. If the goal was not reached, IDA_STAR sets the threshold to that smallest value and repeats.
6. The loop ends when the goal is found (the path is returned) or when no node is cut off (INFINITY), meaning there is no solution.

Properties

Complete: yes, if the branching factor is finite and every step cost is positive.
Optimal: yes, if h(n) is admissible (never over-estimates the real cost).
Space: O(b x d), because only the current path is stored. A* needs to store all generated nodes.
Time: nodes near the start are re-expanded in every round, but this extra work is small compared with the work in the last round.

Result

The pseudocode of IDA* search has been written. It finds an optimal path using the memory of depth first search and the heuristic of A*.